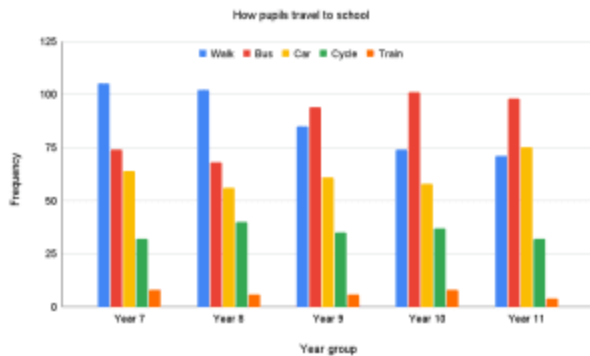




Statistical problems - drawing conclusions

1 Which is the least popular mode of transport in this school? Tick **1** correct answer



☐ bus

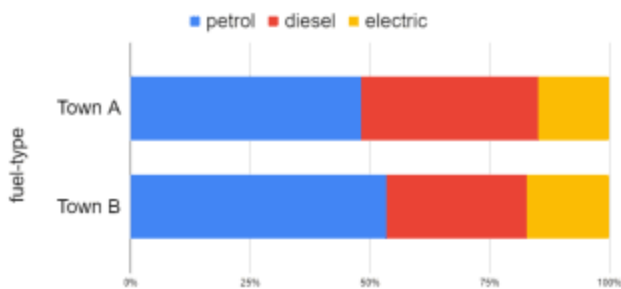
☐ car

☐ cycle

☒ train

☐ walk

2 An investigation surveys residents of two towns about which type of car they drive. The stacked bar chart shows the results. Which is the most popular type of car in this survey? Tick **1** correct answer



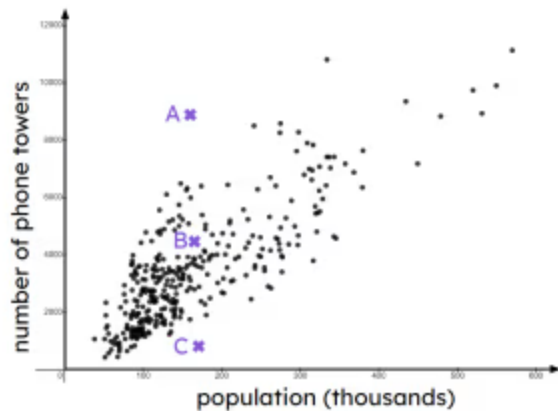
☐ diesel

☐ electric

☒ petrol



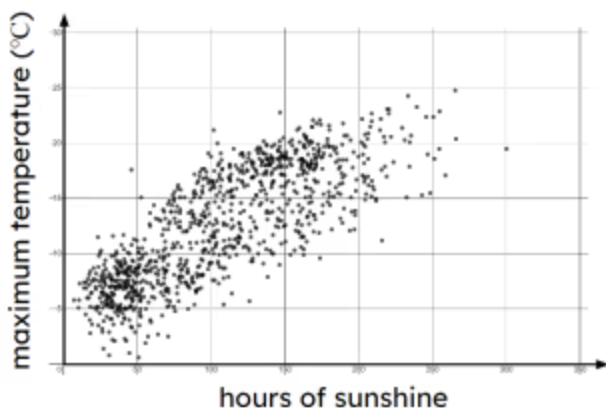
- 3 The scatter graph shows the population of some towns and the number of phone towers the town has. Each point in the scatter graph represents a town or region. Match each town with its statement. Write the correct letter in each box



a	Town A
b	Town B
c	Town C

<i>b</i>	has the expected number of phone towers for the population
<i>c</i>	has a low number of phone towers for the population
<i>a</i>	has a high number of phone towers for the population

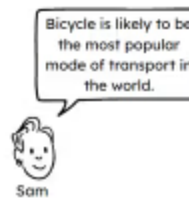
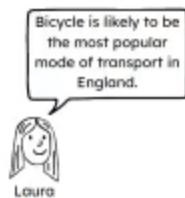
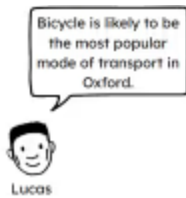
- 4 The scatter graph shows data about the sunshine duration and mean temperature for each month in a city. Which of the statements is the most valid? Tick **1** correct answer



- ☐ A month with 10 hours of sunshine expect a mean temperature of 1 degrees C.
- ☒ A month with 100 hours of sunshine expect a mean temperature of 12 degrees C.
- ☐ A month with 330 hours of sunshine expect a mean temperature of 25 degrees C.



- 5 A data investigation surveys 100 people in Oxford about their preferred mode of transport. The most popular mode of transport was bicycle. Which is the most valid conclusion based on these results? Tick **1** correct answer



- 6 Bias can be present in any sample collected from a population and may affect the results of a statistical investigation (e.g. only asking year 7 pupils what food should be served in the school). Fill in the blank

